WORKSHEET

Chapter 10: Calculations with sample proportions

1 State whether each of the following is a point or interval estimate.

a The variance of heights is about 6.9 cm.

b The median cost of living is between $325 and $455.

c The standard deviation is between 1.6 mg and 1.9 mg.

d The proportion of people with blue eyes is about 34%.

2 From 150 people interviewed in capital cities in Australia, 21 are found to carry a particular gene for a particular type of vision impairment. Estimate the probability that Australians carry the same gene and the variance and standard deviation of the sampling distribution.

3 From 210 tosses, a coin lands tails up 98 times. Estimate the probability of this coin landing with tails up and the variance and standard deviation of the sampling distribution.

4 A total of 105 people who have been accepted as Australian migrants to Australia were surveyed to determine where they intended to permanently reside when they arrive. Forty-two state they will reside in Sydney. Estimate the probability of migrants in Australia residing in Sydney and the variance and standard deviation of the sampling distribution.

© Cengage Learning Australia Pty Ltd 2014

5 A survey of 220 people in Sydney and Melbourne over the age of 18 found that 105 were married (including those in de facto relationships), while the remainder were either divorced, separated, widowed or had never been married. Estimate the probability that an Australian over the age of 18 is married and the variance and standard deviation of the sampling distribution.

6 A survey of 56 people at a shopping centre found that 8 had completed a university degree. Estimate the probability that an Australian has completed a university degree and the variance and standard deviation of the sampling distribution.

7 A survey of 64 people in the Melbourne CBD found that 22 used public transport to commute to work. Estimate the probability that an Australian uses public transport to commute to work and the variance and standard deviation of the sampling distribution.

© Cengage Learning Australia Pty Ltd 2014

Answers

1 **a** Point estimate

b Interval estimate

c Interval estimate

d Point estimate

2 Estimated probability $= \frac{7}{50} = 0.14$; variance $= 0.000\,803$; $\sigma \approx 0.0283$

3 Estimated probability $= \frac{7}{15} \approx 0.47$; variance $\approx 0.002\,07$; $\sigma \approx 0.0455$

4 Estimated probability $= \frac{2}{5} = 0.4$; variance $\approx 0.002\,29$; $\sigma \approx 0.0478$

5 Estimated probability $= \frac{21}{44} \approx 0.477$; variance $\approx 0.001\,13$; $\sigma \approx 0.0337$

6 Estimated probability $= \frac{1}{7} \approx 0.143$; variance $\approx 0.002\,19$; $\sigma \approx 0.0468$

7 Estimated probability $= \frac{11}{32} \approx 0.344$; variance $\approx 0.003\,53$; $\sigma \approx 0.0594$

© Cengage Learning Australia Pty Ltd 2014